

Multimodal RAG for Leishmaniasis Diagnosis

Traceable evidence, model-dependent retrieval, and safety-gated demonstration

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Leishmaniasis is not simple image classification

The Problem

- Overlapping CL, MCL, VL, PKDL, and non-leishmaniasis mimics; lab confirmation is often the decisive evidence.

Evidence Streams

Text

Symptoms

Image

Context

Lab

Weak reference

Why It Is Hard

- Case reports mix symptoms, captions, pathology, treatment, and author conclusions. Image appearance alone is insufficient.

SUPPORT IMAGE

PMC3140149 · PKDL/HIV case report



Infiltrated plaques on nose, wrist, and finger — useful visual context, not standalone ground truth.

Source: open-access PMC case-report image

Four questions, one thread: does evidence help safely?

03

Each RQ asks whether more evidence improves the pipeline without creating unsupported claims.

RQ1

Weak-Reference Evaluation

Can structured case reconstruction support scored diagnosis evaluation on held-out leishmaniasis cases?

RQ2

Small-Model Practicality

Can local open models run on a single GPU with enough latency and memory transparency for audit?

RQ3

Multimodal Evidence

Do images and image-linked context add useful signal, or mainly increase visual hallucination risk?

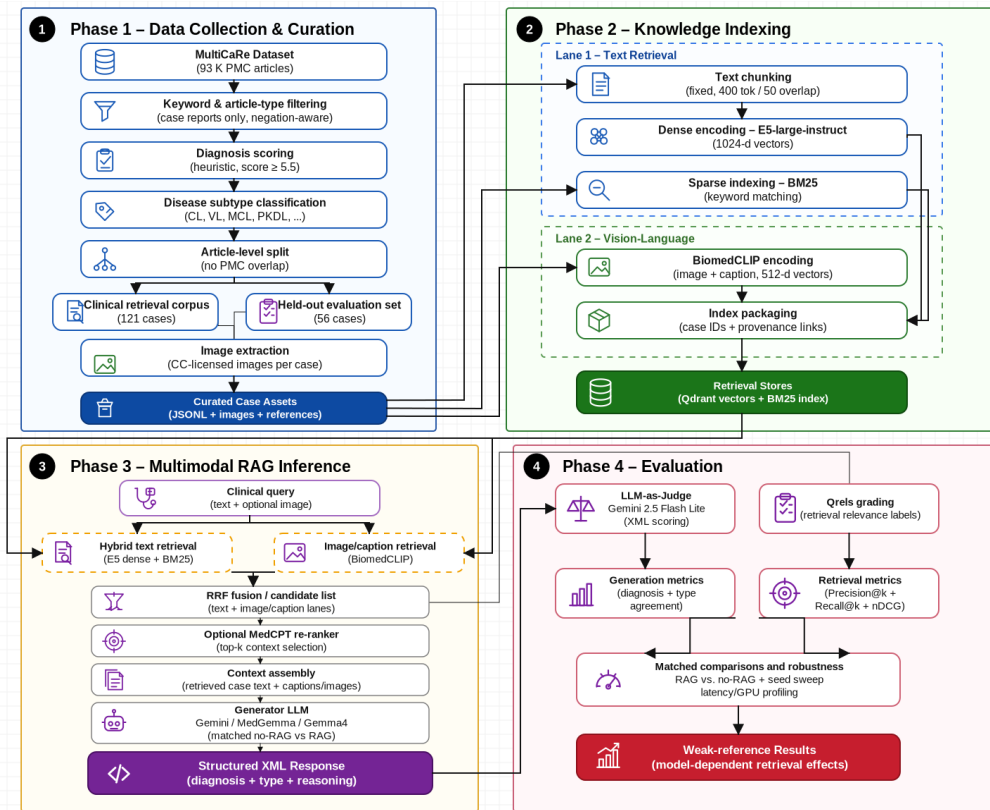
RQ4

Retrieval-Augmented Reasoning

Does retrieving similar cases help generation compared with matched no-RAG conditions?

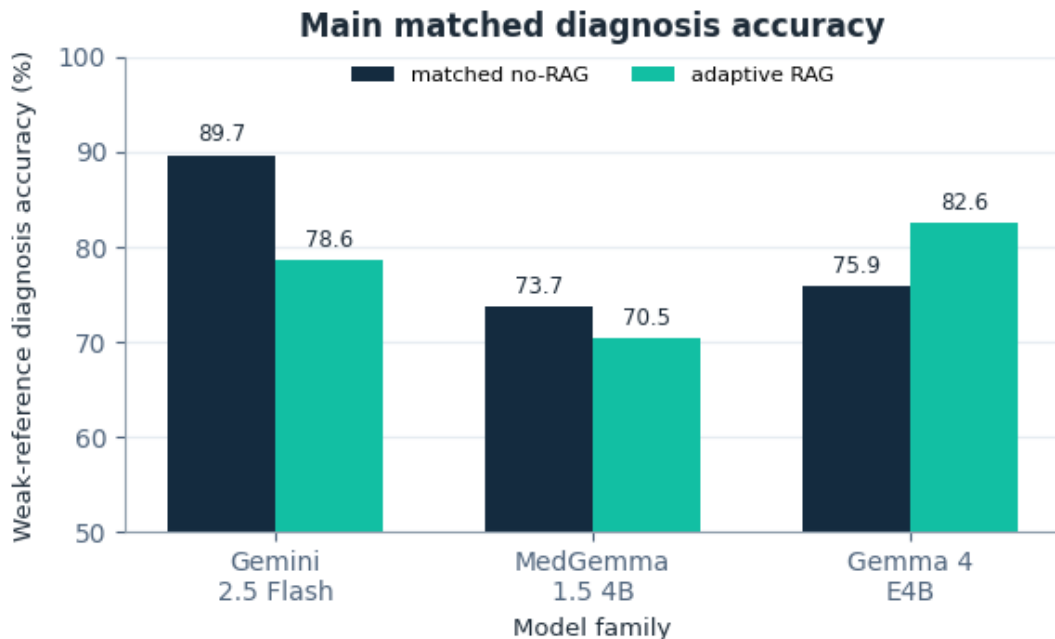
Two-lane retrieval architecture

Text retrieval and image-linked retrieval are fused, gated, and audited before presentation.



Main result: RAG effect is model-dependent

Weak-reference, prompt-judged diagnosis accuracy on 56 held-out cases; x-axis = model family, y-axis = diagnosis accuracy (%).



Key deltas

Gemini **−11.2 pp**

statistically supported degradation

MedGemma **−3.1 pp**

directional; not robust

Gemma 4 **+6.7 pp**

directional; CI crosses 0

Gemma 4 + images **+1.8 pp**

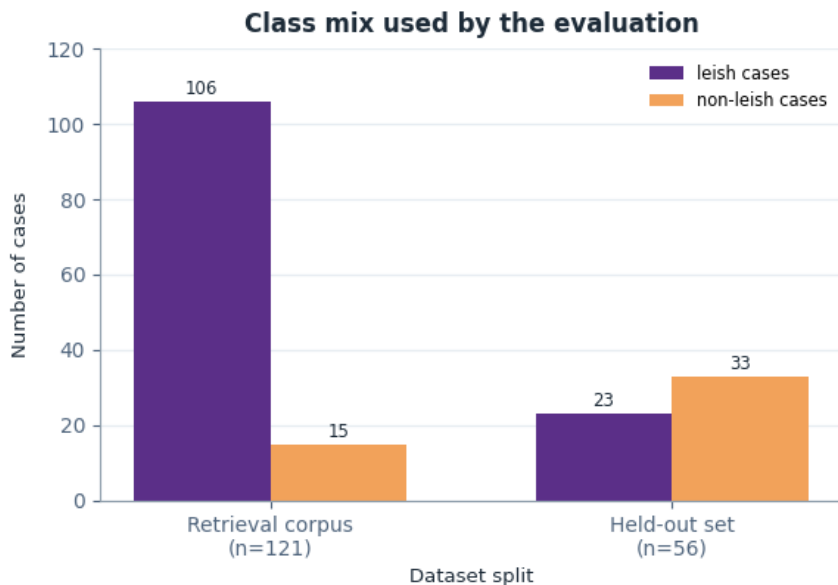
exploratory; hallucination risk ↑

Chart: grouped bars compare matched no-RAG vs adaptive RAG for each model family. Retrieval is not uniformly beneficial.

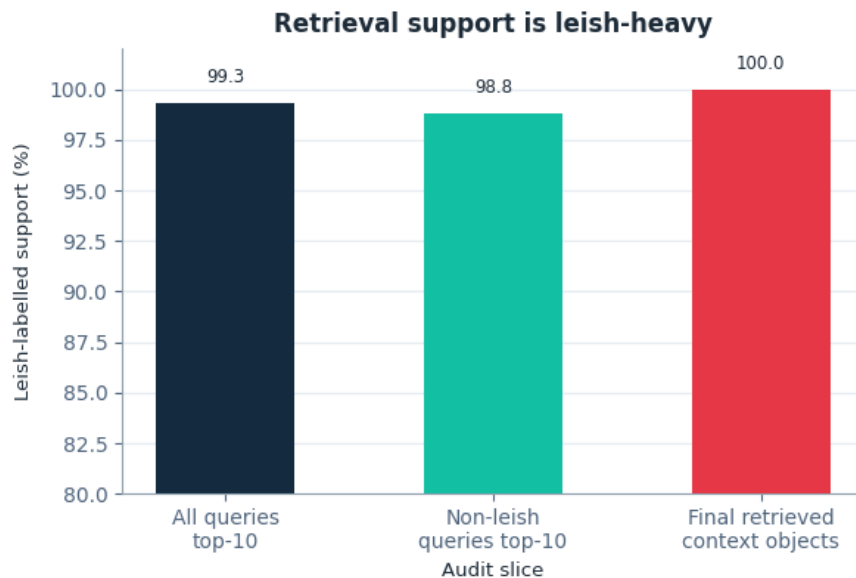
Corpus imbalance explains the specificity risk

Left chart: class mix by split. Right chart: leish-labelled share of retrieved support after clean-context audit.

Retrieval corpus



Held-out / support audit



Key finding: the corpus is 106/121 leish cases while the held-out set is majority non-leish. Non-leish queries still receive mostly leish-labelled top-10 support, so RAG can over-support leishmaniasis mimics.

Retriever quality sets the ceiling

Classic retrieval metrics on 23 held-out queries with qrels; 121-case corpus with 104-doc qrels coverage.

58.3%

P@5

Top-5 precision

59.5%

nDCG@5

Ranking quality @5

71.5%

MRR

First relevant hit

13.5%

Recall@5

Coverage in top 5

23.4%

Recall@10

Coverage in top 10

26.1%

MAP

Overall ranking

Top-rank precision and first-hit quality support explanations, but low recall means relevant cases often fall outside the prompt context window.

Image evidence: useful but risky

Chart axes now show model family (x-axis) and unsupported visual claims as a percentage (y-axis).

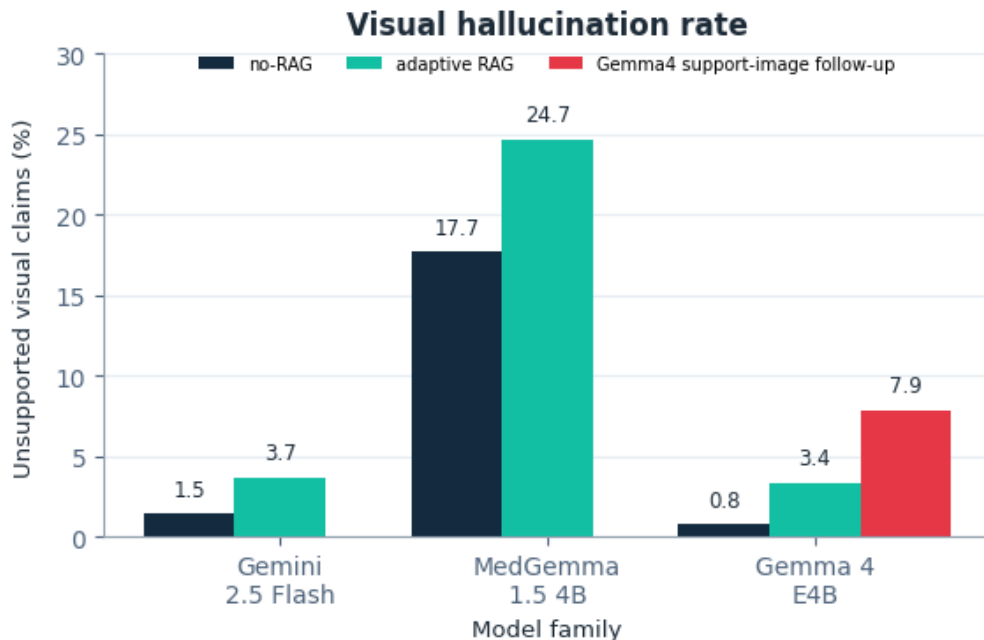


Image-evidence findings

- ✓ Images are context, not ground truth: captions, article text, and lab evidence still carry the diagnosis.
- ✗ PMC3140149 is a support/retrieval example, not a direct official test query.
- ✓ Gemma 4 support-image follow-up was directionally stronger than standard RAG, but the safety cost is visible in the red bar.
- ! Unsupported visual claims rose from 3.4% to 7.9% for the Gemma4 support-image follow-up.
- ✗ Adding images does not automatically improve diagnosis across models.

Visual hallucination = judge-detected unsupported visual/pathology claims. This is not expert adjudication, so the safe demo shows provenance and gate decisions.

Small-model practicality for a live defense

Single 24 GB GPU local models vs. Gemini API: latency, memory, auditability, and demo reliability.

Gemini 2.5 Flash Lite

API · external

Latency: 3–4 s/case
Memory: API/cloud
Best diag. acc.: 89.7%

Strengths

- ✓ Fastest inference
- ✓ Highest no-RAG accuracy
- ✓ Strong multimodal

Limitations

- ✗ RAG degraded under matched evaluation
- ✗ No local GPU audit trail
- ✗ External dependency

MedGemma 1.5 4B

Local · medical-tuned

Latency: 22–26 s/case
Memory: ~16 GB GPU
Best diag. acc.: 73.7%

Strengths

- ✓ Medical domain tuning
- ✓ Full auditability
- ✓ Privacy-safe

Limitations

- ✗ High visual-hallucination rate
- ✗ RAG directionally worsened
- ✗ Weaker reasoning in this setting

Gemma 4 E4B

Local · open

Latency: 46–59 s/case
Memory: ~18–19 GB GPU
Best diag. acc.: 82.6% (RAG)

Strengths

- ✓ Best directional RAG delta (+6.7 pp)
- ✓ Real GPU sidecar demo available
- ✓ Auditable

Limitations

- ✗ Slowest inference
- ✗ Directional, not conclusive
- ✗ Image evidence raises hallucination risk

Limitations are part of the contribution

The thesis claims are deliberately narrow: traceable research evaluation, not bedside-use evidence.

⚠ Weak-reference labels

No clinician adjudication; prompt-judged diagnosis accuracy is useful for controlled comparison, not clinical validation.

⚠ Corpus imbalance

The retrieval corpus is leishmaniasis-heavy, so non-leishmaniasis mimics remain a specificity stress test.

⚠ Small held-out set

Only Gemini's negative RAG delta is statistically supported; Gemma 4's +6.7 pp is directional only.

⚠ Adaptive RAG is route-mixed

Some rows received retrieved context and others fell back. Route-level audit is needed before interpreting aggregate accuracy.

Future: clinician-reviewed labels · balanced mimic set · stronger recall · stricter visual-claim gating · deployment-grade validation

Conclusion: feasible, traceable, not deployment-ready

RQ1 Feasibility

Research pipeline works

Structured cases, retrieval, generation, gate audit, and live demo are reproducible and traceable.

RQ2 Small Models

Practical but bounded

Local models run on a 24 GB GPU, but latency and memory make full live evaluation risky.

RQ3 Multimodal

Useful, risky signal

Image-linked evidence can help, but unsupported visual claims require conservative gating.

RQ4 RAG Effect

Model-dependent

Gemma 4 shows a positive directional RAG pattern; Gemini and MedGemma warn against universal RAG claims.

Final claim: multimodal RAG for leishmaniasis can be evaluated traceably under constrained resources; stronger claims require balanced data, stronger labels, and expert review.

Core result map: evidence behind the main claims

The spoken deck is intentionally compact; this map links headline claims to the supporting thesis results.

Main table set

5.1 main metrics; 5.2-5.3 clean-context; 5.5 modality; 5.6 retrieval; 5.7 route audit; 5.8 imbalance; 5.9 matched deltas.

Why the spoken deck is compressed

The spoken deck compresses many result families into slides 5-10 to preserve a 30-minute defense flow.

What the backup map adds

Labeled charts plus three backup slides for result provenance, clean-context sensitivity, routing, modality, latency, and GPU evidence.

Claim boundary

The backup slides explain existing thesis evidence; they do not create new experimental claims.

Use this map when supervisors ask how each result family connects to the thesis results chapter and appendix.

Clean-context and route audit explain the RAG effect

These mechanism checks explain why the observed RAG effect is conditional.

Clean-context deltas

Gemini -7.1 pp; MedGemma +6.7 pp; Gemma4 +16.1 pp.
Same 56 case IDs, sanitized query artifact; supplementary sensitivity only.

Prompt2 A+B tier

47 cases: Gemini -4.3 pp; MedGemma +4.3 pp; Gemma4 +14.9 pp. Descriptive, not confirmatory.

Route-mixed RAG

Gemma4 audit: 31/56 adaptive_rag rows and 25/56 no-context fallback rows. RAG did not inject context into every case.

Retrieval imbalance

Corpus 106/121 leish; held-out 33/56 non-leish. Non-leish top-10 support was still 98.8% leish-labelled.

Takeaway: retrieval helps conditionally because generator behavior, route policy, and corpus balance all matter.

Sensitivity: modality, latency, and GPU practicality

This slide collects appendix evidence that supports the defense narrative without overclaiming deployment readiness.

Modality matrix

Gemini strongest multimodal no-RAG: 89.7%. Gemma4 text RAG: 85.3%; multimodal RAG: 82.6%; tensor-on RAG: 84.4% supplementary.

Image-risk trade-off

Gemma4 support images: +1.8 pp over standard RAG but visual hallucination rose 3.4% -> 7.9%. Multi-seed +8.3 pp vs no-RAG, CI crosses 0.

Local practicality

MedGemma: 22-26 s/case, peak 15.70 GB. Gemma4: 46-59 s/case, peak 18.84 GB. Gemini API: 3-4 s/case.

Defense point

The contribution is traceability, auditability, and bounded evidence, not a deployment-ready classifier.

These result families support the defense narrative without implying deployment readiness.

Live Demo Case Inspection

CASE 1: PMC7516301_01**Silver: MCL | Live Rank 1: MCL****CASE 2: PMC7456484_01****Silver: PKDL | Live Rank 1: MCL****CASE 3: PMC10026180_04****Label conflict: verified Non-Leish / pseudolabel CL | Live Rank 1: CL**

Case Inspection Setup

Three selected queries from the 56-case held-out set; illustrative, not an accuracy sample
Verified absent from the official 121-case experimental retrieval corpus
Fresh Gemma 4 live-demo outputs used the local defense demo KB on NVIDIA TITAN RTX (4-bit)
Silver labels are pipeline references; aggregate thesis metrics remain the benchmark

Selected Example Outcomes

Case 1 (MCL): live rank-1 type matches the silver reference
Case 2 (PKDL): live rank-1 stays in the leishmaniasis family but shifts to MCL
Case 3: verified Non-Leish / pseudolabel CL; live output stays leishmaniasis-family

Key Findings

Inspectable input-output-evidence traces under local GPU constraints
Case 2 exposes subtype-resolution limits
Case 3 exposes label-conflict and evidence-attribution limits
These cases do not establish clinical validity or deployment readiness
Images provide context, not diagnosis; full inputs: slides 20-22

Case 1: PMC7516301_01 — Concordant Example

SILVER REFERENCE LABEL

**Mucocutaneous
Leishmaniasis (MCL)**

MODEL RANK 1

MCL

**CONFIDENCE
Medium**

**TIME
78.3s**

Clinical Input (Blinded)

19-year-old man from Syria with 6-month throat irritation progressing to severe throat pain and fever; exam noted prior arm scar and oropharyngeal/laryngeal mucosal lesions.

Held-out case absent from the official 121-case experimental retrieval corpus; full text is on slide 20.

Fresh live output used the local defense demo KB; target and confirmatory fields were withheld.



Input image

Model Output Summary

Top diagnosis: Mucocutaneous Leishmaniasis (Mucosal involvement)

Runtime support: 3 retrieved chunks

Output route: retrieval-supported response

LLM Council Review Synthesis

Review synthesis; not Gemma output

Reviews agree MCL Rank 1 is a strong concordant positive example
Image is visually consistent/contextual, not diagnostic proof

Runtime Support Returned by Demo KB

1. Mucocutaneous leishmaniasis: runtime support returned by the backend
2. Cutaneous leishmaniasis: runtime support returned by the backend

Case 2: PMC7456484_01 — Subtype Mismatch

SILVER REFERENCE LABEL

**Post-Kala-Azar Dermal
Leishmaniasis (PKDL)**

MODEL RANK 1

MCL

**CONFIDENCE
Medium**

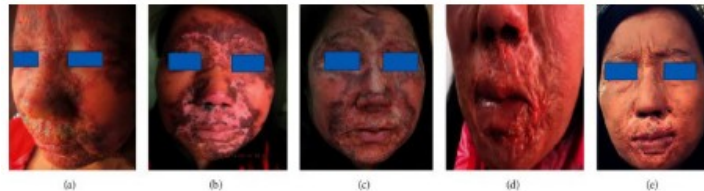
**TIME
74.0s**

Clinical Input (Blinded)

37-year-old woman from eastern Bhutan with extensive facial plaques, eyelid damage/scarring, nasal mucosal extension, and prior treated visceral leishmaniasis history.

Held-out case absent from the official 121-case experimental retrieval corpus; full text is on slide 21.

Fresh live output used the local defense demo KB; target and confirmatory fields were withheld.



Input image

Model Output Summary

Top diagnosis: Chronic Mucocutaneous Leishmaniasis
(Mucocutaneous Leishmaniasis)

Runtime support: 3 retrieved chunks

Output route: retrieval-supported response

LLM Council Review Synthesis

Review synthesis; not Gemma output

Reviews agree the family is retained but PKDL is not named
Demo KB lacks a PKDL anchor; subtype limitation

Runtime Support Returned by Demo KB

1. Visceral leishmaniasis: runtime support returned by the backend
2. Mucocutaneous leishmaniasis: runtime support returned by the backend

Case 3: PMC10026180_04 — Label-Conflict Stress Test

LABEL CONFLICT

**Verified Non-Leish /
pseudolabel CL**

MODEL RANK 1

CL

**CONFIDENCE
Medium**

**TIME
61.5s**

Clinical Input (Blinded)

11-year-old girl with ataxia telangiectasia, recurrent respiratory and ear infections, low IgG, and a spreading nose rash extending to the nostrils and left cheek; the image/text are leishmaniasis-plausible but the pipeline labels conflict.

Held-out case absent from the official 121-case experimental retrieval corpus; full text is on slide 22.

Fresh live output used the local defense demo KB; target and confirmatory fields were withheld.



Input image

Model Output Summary

Top diagnosis: Mucocutaneous Leishmaniasis (or other atypical infection)

Runtime support: 3 retrieved chunks

Output route: retrieval-supported response

LLM Council Review Synthesis

Review synthesis; not Gemma output

Reviews flag a label conflict; full-text case is leishmaniasis-plausible

Live CL/MCL-family output is not specificity proof; confirmation required

Runtime Support Returned by Demo KB

1. Cutaneous leishmaniasis: runtime support returned by the backend
2. Cutaneous leishmaniasis: runtime support returned by the backend



Thank You

Questions?

Feasible and traceable under constraints; not deployment-ready.

Multimodal RAG for Leishmaniasis Diagnosis

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Full Blinded Input: PMC7516301_01

CASE ID

PMC7516301_01

INPUT MODE

Text + image

SOURCE

Held-out query

USE

Q&A appendix

Full blinded clinical input

A 19-year-old man from Syria (arrived in Denmark in 2015) presented in June 2016 with a 6-month history of throat irritation that progressed over several days to severe throat pain and fever. He had no history of severe illness or excess infections; examination noted a scar on his arm that was suggestive of previous cutaneous leishmaniasis. Laryngoscopy demonstrated massive edema of the mucosa of both the pharynx and larynx; the epiglottis and uvula were swollen and red with papilloma-like lesions. Endoscopic images show marked swelling and papillomatous-appearing mucosal lesions of the oropharynx and larynx.



Input image used in test

Full blinded clinical input: diagnosis, title, abstract, captions, and confirmatory findings withheld.

Live case inspection - fresh local-GPU demo output

Full Blinded Input: PMC7456484_01

CASE ID

PMC7456484_01

INPUT MODE

Text + image

SOURCE

Held-out query

USE

Q&A appendix

Full blinded clinical input

A 37-year-old woman from eastern Bhutan presented in 2014 with extensive erythematous plaques on the forehead, central face, perioral area, and cheeks extending to both ears. The left eye was damaged and there was marked swelling of the right eyelid with lid scarring and lagophthalmos. Plaques extended to the nasal mucosa without intraoral involvement. Other skin and systemic examinations were reportedly normal.

Laboratory testing showed hemoglobin 11.7 g/dl, total leukocyte count 6,390/mm³ (neutrophils 62%, lymphocytes 28%), platelets 191,000/mm³, ESR 39 mm in the 1st hour, SGOT 35 U/L, SGPT 68 U/L, bilirubin 1 mg/dl, urea 21 mg/dl, creatinine 0.8 mg/dl, fasting glucose 80 mg/dl. Tests for HIV, hepatitis B and C, and syphilis were negative. Chest radiograph, ECG, and abdominal ultrasound were normal.

Full blinded clinical input (continued)

Relevant past history: in February 1999, during pregnancy she had fever, weight loss, anemia, and hepatomegaly and “was diagnosed as VL and treated with SSG injections 850 mg/day for 20 days.” Seven months after that illness she developed erythematous lesions on her forehead that spread over the face. In 2002 a dermatologist recorded papules and plaques over the cheeks, periorbital, and perioral areas that “were suspected as chronic dermatitis or cutaneous lupus erythematosus,” and she received topical, intralesional, and oral corticosteroids. In 2007 a WHO consultant for kala azar “suspected PKDL.” Skin biopsies performed in 2006, 2010, and 2013 “failed to confirm leishmaniasis,” and no firm diagnosis had been recorded.

Full blinded clinical input: diagnosis, title, abstract, captions, and confirmatory findings withheld.

Live case inspection - fresh local-GPU demo output

Full Blinded Input: PMC10026180_04

CASE ID

PMC10026180_04

INPUT MODE

Text + image

SOURCE

Held-out query

USE

Q&A appendix

Full blinded clinical input

An 11-year-old Arab girl with a known diagnosis of ataxia telangiectasia (diagnosed at 1 year) presented with a spreading skin rash on her nose that extended to the nostrils and her left cheek. Family history included consanguineous parents and another sibling with AT. Her childhood course included ataxia, recurrent lung and ear infections, sclerosing cholangitis, hypothyroidism, and later diffuse large B-cell lymphoma. She used a wheelchair and had multiple café-au-lait spots and telangiectasias. Immunologic testing showed a decreased immunoglobulin level (IgG 516 mg/L) with lymphocyte count and function reported as normal. Her baseline therapies included monthly IVIG, inhaled budesonide and fluticasone, and levothyroxine.

Initially she received amoxicillin-clavulanic acid and topical chloramphenicol without clinical response. On examination, the tip of the nose had an erythematous plaque with a central erosion and a circinate-elevated border with overlying scattered pustules; the nose bridge and nostrils had small erythematous erosions. An atypical infection was considered.



Input image used in test

Full blinded clinical input: diagnosis, title, abstract, captions, and confirmatory findings withheld.

Live case inspection - fresh local-GPU demo output

Gemma 4 RAG/No-RAG Backup: PMC7516301_01

CASE ID	SILVER REFERENCE	USE	SOURCE
PMC7516301_01	MCL	Q&A backup only	Gemma 4 pipeline

Source: official Gemma 4 experiment-pipeline comparison; same held-out case ID, Q1/Q3 multimodal query type.

NO-RAG condition

Rank 1 type: Non-Leishmaniasis
Rank 1: Non-Leishmaniasis
Confidence: High
Retrieved context chunks: 0
Experiment pipeline: no retrieved context

RAG condition

Rank 1 type: MCL
Rank 1: Mucosal Leishmaniasis (Mucocutaneous Leishmaniasis)
Confidence: High
Retrieved context chunks: 3
Experiment pipeline: retrieved context used

Presentation interpretation

Retrieval changed this selected backup example from Non-Leishmaniasis to MCL in the official Gemma 4 experiment-pipeline comparison. Selected case-level backup only; aggregate thesis metrics remain the benchmark. Not clinical validation and not diagnosis from image alone.

Gemma 4 RAG/No-RAG Backup: PMC7456484_01

CASE ID PMC7456484_01	SILVER REFERENCE PKDL	USE Q&A backup only	SOURCE Gemma 4 pipeline
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Source: official Gemma 4 experiment-pipeline comparison; same held-out case ID, Q1/Q3 multimodal query type.

NO-RAG condition

Rank 1 type: MCL
Rank 1: Mucocutaneous Leishmaniasis (MCL)
Confidence: High
Retrieved context chunks: 0
Experiment pipeline: no retrieved context

RAG condition

Rank 1 type: MCL
Rank 1: Mucocutaneous Leishmaniasis (Mucocutaneous form)
Confidence: High
Retrieved context chunks: 1
Experiment pipeline: retrieved context used

Presentation interpretation

Both conditions remain inside the leishmaniasis family but miss the PKDL subtype, so this remains a subtype-resolution challenge.
Selected case-level backup only; aggregate thesis metrics remain the benchmark.
Not clinical validation and not diagnosis from image alone.

Gemma 4 RAG/No-RAG Backup: PMC10026180_04

CASE ID	SILVER REFERENCE	USE	SOURCE
PMC10026180_04	verified Non-Leish / pseudolabel CL	Q&A backup only	Gemma 4 pipeline

Source: official Gemma 4 experiment-pipeline comparison; same held-out case ID, Q1/Q3 multimodal query type.

NO-RAG condition

Rank 1 type: Non-Leishmaniasis
Rank 1: Mucocutaneous infection/Inflammatory dermatosis secondary to underlying i...
Confidence: High
Retrieved context chunks: 0
Experiment pipeline: no retrieved context

RAG condition

Rank 1 type: Non-Leishmaniasis
Rank 1: Mucocutaneous infection/Inflammatory dermatosis secondary to underlying i...
Confidence: High
Retrieved context chunks: 0
Experiment pipeline: retrieved context used

Presentation interpretation

The official comparison returns Non-Leishmaniasis in both conditions, but the full-text live demo and LLM council reviews make this a label-conflict/evidence-attribution stress test, not specificity proof.
Selected case-level backup only; aggregate thesis metrics remain the benchmark.
Not clinical validation and not diagnosis from image alone.

Official RAG Trace: PMC7516301_01

CASE ID PMC7516301_01	RETRIEVER hybrid	RERANK True	TOP K 20	USE Q&A trace only
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qid: PMC7516301_01::Q1_Q3_multimodal_diagnosis | source: official Gemma 4 experiment-pipeline trace

Visible contexts are the rerank-enabled final context list used for generation; no separate pre-rerank list is claimed.

Context rank 1

doc_id: PMC7528117_01
score: 0.03301807008521583
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 52-year-old woman who works in sugarcane fields in a rural area was referred to Oral Medicine for evaluation of a granulomatous lesion on the palate and for hoarseness that she thought followed a flu 3 months earlier. According

Context rank 2

doc_id: PMC7527865_05
score: 0.01497737556561086
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 30-year-old Caucasian man, unemployed, HIV-negative and with no other concomitant infections, presented in 2014 with a papule on the inferior eyelid of the right eye. His past history included, in 2008, an isolated nodulation in

Context rank 3

doc_id: PMC7585787_01
score: 0.014164989939637825
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 38-year-old man from Yayu district, western Ethiopia, presented to the outpatient clinic with multiple skin lesions on his face and hands and mutilation of the nose; his upper lip was injured and swollen. Eight years earlier he

Fresh Real-GPU Output Audit: PMC7516301_01

CASE ID

PMC7516301_01

PROVIDER

real_gpu_gemma4

MODEL

google/gemma-4-E4B-it

USE

Fresh audit only

request_id: v12d_recapture_live_gpu_PMC7516301_01_20260617_165636 | source: /home/ngocnt/Leishmaniasis_v3/rag/instructions/process/14/demo/presentation/v12d/re-capture/20260617_165636/PMC7516301_01_live_gpu_result.json

Exact model-output fields from fresh backend response

Rank 1 supportive consideration: Mucocutaneous Leishmaniasis (Mucosal involvement)

Rank 1 Diagnosis Type: MCL

Rank 1 Confidence: Medium

safety_state: generated_support | elapsed_seconds: 78.336 | query_image_tensor_count: 1

Exact returned evidence chunks from raw backend response

Evidence rank 1

chunk_id: mcl-002 | score: 0.3617
title: Palatal and nasal mucosal disease in leishmaniasis
diagnosis_label: Mucocutaneous leishmaniasis | confirmatory: True
excerpt: Palatal or nasal mucosal lesions with compatible epidemiology should raise concern for mucocutaneous leishmaniasis and prompt tissue confirmation.

Evidence rank 2

chunk_id: cl-001 | score: 0.3038
title: Cutaneous leishmaniasis with ulcer and smear-confirmed amastigotes
diagnosis_label: Cutaneous leishmaniasis | confirmatory: True
excerpt: A patient with an ulcerated plaque on the forearm after sandfly exposure had a lesion smear showing amastigotes. The presentation was consistent with localized cutaneous leishmaniasis.

Evidence rank 3

chunk_id: vl-001 | score: 0.2457
title: Visceral leishmaniasis with prolonged fever and splenomegaly
diagnosis_label: Visceral leishmaniasis | confirmatory: True
excerpt: Prolonged fever, weight loss, pancytopenia, and splenomegaly are classic features of visceral leishmaniasis. Bone marrow or tissue confirmation is typically required.

Official RAG Trace: PMC7456484_01

CASE ID PMC7456484_01	RETRIEVER hybrid	RERANK True	TOP K 20	USE Q&A trace only
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qid: PMC7456484_01::Q1_Q3_multimodal_diagnosis | source: official Gemma 4 experiment-pipeline trace

Visible contexts are the rerank-enabled final context list used for generation; no separate pre-rerank list is claimed.

Context rank 1

doc_id: PMC7585787_01
score: 0.01971861471861472
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 38-year-old man from Yayu district, western Ethiopia, presented to the outpatient clinic with multiple skin lesions on his face and hands and mutilation of the nose; his upper lip was injured and swollen. Eight years earlier he

Context rank 2

doc_id: PMC6019596_01
score: 0.019506936428773658
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 78-year-old man from a rural area in western Iran was referred for multiple ulcerative, exudative lesions involving the mid-face (upper lip and anterior nasal fossa), above the eyebrows, dorsal aspects of both hands, and posteri

Context rank 3

doc_id: PMC4934146_02
score: 0.0151288056206089
diagnosis_type: DsCL | label_source: train_verified
text_prefix: A 40-year-old woman from Balao Chico (Guayas province, Ecuador) presented in August 1979 with a progressive widespread skin eruption. The illness began with a lesion on the lumbar region; papular lesions developed and became pustul

Fresh Real-GPU Output Audit: PMC7456484_01

CASE ID
PMC7456484_01

PROVIDER
real_gpu_gemma4

MODEL
google/gemma-4-E4B-it

USE
Fresh audit only

request_id: v12d_recapture_live_gpu_PMC7456484_01_20260617_165636 | source: /home/ngocnt/Leishmaniasis_v3/rag/instructions/process/14/demo/presentation/v12d/re-capture/20260617_165636/PMC7456484_01_live_gpu_result.json

Exact model-output fields from fresh backend response

Rank 1 supportive consideration: Chronic Mucocutaneous Leishmaniasis (Mucocutaneous Leishmaniasis)

Rank 1 Diagnosis Type: MCL

Rank 1 Confidence: Medium

safety_state: generated_support | elapsed_seconds: 73.986 | query_image_tensor_count: 1

Exact returned evidence chunks from raw backend response

Evidence rank 1

chunk_id: vl-002 | score: 0.5146
title: Kala-azar pattern with hepatosplenomegaly
diagnosis_label: Visceral leishmaniasis | confirmatory: False
excerpt: Kala-azar often presents with hepatosplenomegaly, chronic fever, anemia, and systemic illness rather than an isolated skin ulcer.

Evidence rank 2

chunk_id: mcl-001 | score: 0.3856
title: Mucocutaneous leishmaniasis with nasal and oral involvement
diagnosis_label: Mucocutaneous leishmaniasis | confirmatory: False
excerpt: Nasal septal destruction, oral ulceration, and mucosal inflammation after prior cutaneous disease support mucocutaneous leishmaniasis.

Evidence rank 3

chunk_id: cl-002 | score: 0.3795
title: Papulonodular skin lesion in endemic travel history
diagnosis_label: Cutaneous leishmaniasis | confirmatory: True
excerpt: A chronic papule progressing to a nodular ulcer after endemic travel supported cutaneous leishmaniasis. Biopsy and PCR were used for confirmation.

Official RAG Trace: PMC10026180_04

CASE ID PMC10026180_04	RETRIEVER hybrid	RERANK True	TOP K 20	USE Q&A trace only
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qid: PMC10026180_04::Q1_Q3_multimodal_diagnosis | source: official Gemma 4 experiment-pipeline trace

Visible contexts are the rerank-enabled final context list used for generation; no separate pre-rerank list is claimed.

Context rank 1

doc_id: PMC5490479_01
score: 0.021050130672368734
diagnosis_type: LR | label_source: train_verified
text_prefix: A 9-year-old boy presented with multiple papular lesions on the left cheek that had been present for three months and had gradually enlarged. His medical history was notable for an episode of cutaneous leishmaniasis located on the

Context rank 2

doc_id: PMC3523429_01
score: 0.015682382133995035
diagnosis_type: CL | label_source: train_verified
text_prefix: A 52-year-old woman presented with a single lesion on the tip of her nose that had existed for 14 months. It began as a papule and was slowly increasing in size, enlarging to a plaque. A smear of the lesion performed about 1 year

Context rank 3

doc_id: PMC7585787_01
score: 0.014718614718614718
diagnosis_type: MCL | label_source: train_verified
text_prefix: A 38-year-old man from Yayu district, western Ethiopia, presented to the outpatient clinic with multiple skin lesions on his face and hands and mutilation of the nose; his upper lip was injured and swollen. Eight years earlier he

Fresh Real-GPU Output Audit: PMC10026180_04

CASE ID

PMC10026180_04

PROVIDER

real_gpu_gemma4

MODEL

google/gemma-4-E4B-it

USE

Fresh audit only

request_id: v12d_recapture_live_gpu_PMC10026180_04_20260617_165636 | source: /home/ngocnt/Leishmaniasis_v3/rag/instructions/process/14/demo/presentation/v12d/re-capture/20260617_165636/PMC10026180_04_live_gpu_result.json

Exact model-output fields from fresh backend response

Rank 1 supportive consideration: Mucocutaneous Leishmaniasis (or other atypical infection)

Rank 1 Diagnosis Type: CL

Rank 1 Confidence: Medium

safety_state: generated_support | elapsed_seconds: 61.477 | query_image_tensor_count: 1

Exact returned evidence chunks from raw backend response

Evidence rank 1

chunk_id: cl-002 | score: 0.3079

title: Papulonodular skin lesion in endemic travel history

diagnosis_label: Cutaneous leishmaniasis | confirmatory: True

excerpt: A chronic papule progressing to a nodular ulcer after endemic travel supported cutaneous leishmaniasis. Biopsy and PCR were used for confirmation.

Evidence rank 2

chunk_id: cl-001 | score: 0.2318

title: Cutaneous leishmaniasis with ulcer and smear-confirmed amastigotes

diagnosis_label: Cutaneous leishmaniasis | confirmatory: True

excerpt: A patient with an ulcerated plaque on the forearm after sandfly exposure had a lesion smear showing amastigotes. The presentation was consistent with localized cutaneous leishmaniasis.

Evidence rank 3

chunk_id: vl-002 | score: 0.0262

title: Kala-azar pattern with hepatosplenomegaly

diagnosis_label: Visceral leishmaniasis | confirmatory: False

excerpt: Kala-azar often presents with hepatosplenomegaly, chronic fever, anemia, and systemic illness rather than an isolated skin ulcer.